

Structural Change and Manufacturing Productivity Growth in Odisha: Evidence from Organised Industries

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Abstract

Structural change drives the reallocation of labour across sectors and shapes productivity, employment and economic growth. This study examines the patterns and determinants of labour-productivity growth in the manufacturing sector of Odisha using industry-level data from the Annual Survey of Industries (ASI). Its objective is to analyse the contribution of factor accumulation, technological progress and structural change to manufacturing productivity growth across industries. A growth-accounting framework decomposes aggregate value-added growth into the contributions of capital, labour and total factor productivity (TFP), while a structural-decomposition (production-function) approach following de Vries et al. (2015) separates productivity gains arising *within* industries from those arising from the *reallocation* of resources across industries. Gross-output growth is further computed for twenty-three 2-digit NIC manufacturing divisions over 2014–15 to 2023–24. The results show that within-industry improvements and TFP, rather than factor accumulation, are the dominant sources of manufacturing productivity growth in most years, while reallocation effects are usually small but occasionally decisive: in 2022–23 a sharp negative static reallocation (–15.56 percentage points) turned labour-productivity growth negative (–10.97 percentage points) amid post-pandemic reverse structural transformation and energy-price shocks. Gross-output growth was highly uneven across industries, with food products, electrical equipment, fabricated metals, basic metals and chemicals emerging as the principal growth drivers. The findings underline that allocative efficiency and productivity-enhancing policies, alongside investment in physical and human capital, are central to sustaining manufacturing growth in a sub-national developing-economy setting.

Keywords: structural change; labour-productivity decomposition; growth accounting; total factor productivity; gross-output growth; manufacturing; Odisha (ASI).

JEL classification: O14; O47; L60; D24; R11.

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1 Introduction

Sustained economic growth accompanied by structural transformation is a necessary condition for India to achieve its developmental goals over the next few decades. While the Indian economy has consistently been among the fastest-growing large economies over the past few decades, this growth has not been accompanied by adequate structural transformation (Basole, 2022; Ghose, 2020; Majid, 2019). Unlike the East Asian trajectory—characterised by rapid labour absorption into manufacturing together with sustained productivity growth—India’s development path has been marked by persistent dualism.

This configuration has meant that aggregate output and GDP per capita have grown without a commensurate expansion of productive employment. As a result, there have been long-standing concerns about jobless growth, stagnant wages and persistently low female labour-force participation. Since 2018, employment levels have increased, female labour-force participation has risen, and India has emerged from the COVID-19 pandemic as the fastest-growing large economy in the world, combining robust output growth with relatively low inflation (Ministry of Finance, 2026).

Historically, Odisha, with its rich natural resources, has been known primarily for agriculture and mining. In recent years, however, there has been a notable shift towards industry and services, which has reshaped the economic landscape of the state (Government of Odisha, 2022). Notably, Odisha appears to be transitioning from agriculture directly towards industry, somewhat bypassing the service sector: its economy is distinctly more industry-oriented than India’s as a whole. The manufacturing and mining-and-quarrying sectors are particularly developed in Odisha, whereas services are relatively more developed at the national level.

Manufacturing’s key strength lies in its ability to employ low-skilled labour migrating from agriculture and to enhance their productivity substantially. Moreover, the growth of manufacturing stimulates related activities such as construction and services—including distributive trade and transport—which similarly employ low-skilled labour at higher productivity levels. Rapid manufacturing growth therefore leads to a swift transfer of labour from low-productivity agriculture to higher-productivity non-agricultural sectors, further boosting growth and improving employment conditions. This suggests that in low-income and developing economies manufacturing-led growth is typically faster and more feasible, whereas in high-income economies service-led growth tends to accelerate more rapidly. Against this background, this paper quantifies the contribution of factor accumulation, technological progress and structural change to aggregate manufacturing productivity growth in Odisha over 2014–15 to 2023–24, using ASI data. Its contribution is threefold: (i) it provides state-level evidence for Odisha at the 2-digit and 3-digit NIC level; (ii) it applies the production-function decomposition of de Vries et al. (2015) and the direct-aggregation growth-accounting framework of Jorgenson et al. (2005, 2007) to a single sub-national manufacturing sector; and (iii) it documents and interprets a pronounced 2022–23 reallocation anomaly in the post-pandemic period.

2 Literature Review

Structural transformation and employment are interconnected dynamics that play a crucial role in the socio-economic development of countries. Structural transformation refers to the process by which a nation’s economic base shifts from primarily agriculture and low-productivity activities to more industrialised and service-oriented sectors. Structural change has been at the heart of development economics for a long time and has been argued to be an important aspect of economic development.

In the Kuznets framework, economic growth is driven by the reallocation of labour from

agriculture, which has lower productivity, to the non-agricultural sector, which has higher productivity (Kuznets, 1966). Lewis (1954) describes a low-income economy as comprising two sectors: a traditional sector characterised by low productivity and surplus labour, and a smaller modern sector with high productivity and a smaller labour force that expands through capital accumulation. Clark (1940), Kuznets (1966) and Maddison (1980) defined structural change and economic growth in related terms; according to them, structural change linked to economic growth initially involves moving labour from agriculture to both manufacturing and services. Kaldor (1967) emphasised, through his growth laws, that labour productivity grows faster in manufacturing than in agriculture or services, owing to technology spillovers, leading to higher overall economic growth.

For India, the productivity literature is extensive but largely focused on formal manufacturing (Ahluwalia, 1991; Balakrishnan and Pushpangadan, 1994; Goldar, 2004; Das and Kalita, 2011; Goldar, 2014; Kathuria et al., 2014; Goldar, 2015). A growing body of work applies structural-decomposition methods to India and other large emerging economies. de Vries et al. (2012) deconstruct aggregate productivity growth in the BRIC economies and show that the apparent gains from labour reallocation are sensitive to the formal–informal distinction. Erumban et al. (2019) find that static structural change has, on balance, raised aggregate labour productivity in India as workers moved towards higher-productivity sectors. Jorgenson and Timmer (2011) document a new set of stylised facts on structural change in advanced economies, while Jorgenson (1963) provides the foundations of the capital-input and investment framework that underlies modern growth accounting. For India specifically, Jayadev et al. (2025) document a puzzling slowdown in labour productivity alongside rising employment, attributing it to intensifying dualism and a reversal of labour reallocation towards lower-productivity activities—a pattern echoed in the present study’s 2022–23 results.

Methodologically, the present study builds on the labour-productivity decomposition of de Vries et al. (2015), the sectoral growth-accounting tradition of van Ark (1996) and Timmer and Szirmai (2000), and the direct-aggregation approach to industry and aggregate TFP of Jorgenson et al. (2005, 2007). Only a few studies have analysed the manufacturing productivity of the coastal state of Odisha (Mahanta, 2025); the contribution of this paper to the Indian literature is to apply these frameworks at the state level.

3 Data and Methodology

3.1 Data

The analysis uses Annual Survey of Industries (ASI) data published by the Ministry of Statistics and Programme Implementation (Ministry of Statistics and Programme Implementation, 2024). Industry aggregates of real value added, number of workers, capital input, intermediate input and the corresponding factor-income shares are constructed at the 2-digit and 3-digit levels of the National Industrial Classification (NIC-2008) for the manufacturing sector of Odisha. The labour productivity decomposition is computed for the years 2016–17 to 2023–24 and the industry gross-output growth rates for 2014–15 to 2023–24.

3.2 Static and dynamic reallocation effects: decomposition of labour productivity growth

Aggregate economic activity is measured by an aggregate production function. Aggregate value added (V_t^{PF}) is obtained by summing value added across industries, and aggregate employment (L_t^*) as the sum of the number of workers across industries. The aggregate labour-productivity level in year t is then

$$v_t^{PF} = \frac{V_t^{PF}}{L_t^*}. \quad (1)$$

In Equation (1), $V_t^{PF} = \sum_i V_{i,t}$, with $V_{i,t}$ the real value added of industry i ; $L_t^* = \sum_i L_{i,t}$, with $L_{i,t}$ the number of workers in industry i ; and v_t^{PF} is labour productivity, i.e. real value added per worker. Following de Vries et al. (2015), the annual change in the labour-productivity level is decomposed into a within-industry productivity change and a reallocation effect:

$$\Delta v_t^{PF} = \sum_i \Delta v_{i,t} u_{i,t-1} + \sum_i \Delta u_{i,t} v_{i,t-1} + \sum_i \Delta u_{i,t} \Delta v_{i,t}. \quad (2)$$

Here $u_{i,t}$ is the employment share of industry i in aggregate employment, and the operator Δ indicates the change over the previous year, i.e. $\Delta u_{i,t} = u_{i,t} - u_{i,t-1}$. The three terms of Equation (2) have the following interpretation:

- The **first term** is the *within effect*: the effect of productivity change within each industry, i.e. the sum of changes in productivity levels in individual industries.
- The **second term** measures worker reallocation across sectors — the change in employment share weighted by levels of productivity — often called the *static reallocation* (or static between) effect. A positive static reallocation effect indicates worker movement to sectors with above-average productivity levels.
- The **third term** is an interaction of the change in employment share and the change in productivity; it measures the combined effect of changing employment shares and industry productivity — a *dynamic reallocation* term (van Ark, 1996; Timmer and Szirmai, 2000).

3.3 Contribution of factor inputs and TFP growth to aggregate value-added growth

As in the case of labour-productivity growth, the production-function (PF) approach allows aggregate value-added growth (V_t^{PF}) to be decomposed into the contributions of aggregate capital input (K), aggregate labour input (l) and aggregate total factor productivity (A) growth, using the standard growth-accounting method:

$$\Delta V_t^{PF} = \bar{s}_{k,t} \Delta \ln K_t + \bar{s}_{l,t} \Delta \ln l_t + \Delta \ln A_t^{PF}. \quad (3)$$

In Equation (3), s_k is the share of aggregate capital compensation and s_l the share of aggregate labour compensation in aggregate nominal value added, both averaged over the current and previous periods (the over-bars denote this two-period average); $\Delta \ln K_t$ is the aggregate capital-services growth rate and $\Delta \ln l_t$ the aggregate labour-input growth rate; and $\Delta \ln A_t^{PF}$ is the growth of aggregate total factor productivity. Aggregate capital and labour inputs are themselves the sums of industry capital and labour inputs.

Note that the decomposition in Equation (3) uses log differences of output and inputs, whereas the decomposition in Equation (2) uses differences in levels (later converted to percentage changes). The two are therefore not strictly comparable in absolute magnitude.

Jorgenson et al. (2005) use a less restrictive production-possibility-frontier approach, which relaxes the restrictions of the industry value-added function. In this approach the measurement of output differs from the aggregate production function, while the measurement of inputs is unchanged. Defining aggregate value-added growth as a translog index of industry value-added

growth, and keeping capital and labour as in Equation (3), aggregate value-added growth can be expressed as

$$\Delta V_t^{PF} = \sum_i \bar{s}_{i,t} \Delta \ln V_{i,t} = \bar{s}_{k,t} \Delta \ln K_t + \bar{s}_{l,t} \Delta \ln l_t + \Delta \ln A_t^{PF}, \quad (4)$$

where $\Delta \ln V_{i,t}$ is the growth rate of real value added in industry i , and $\bar{s}_{i,t}$ is the average share of industry i in aggregate nominal value added between years $t - 1$ and t , computed as

$$\bar{s}_{i,t} = \frac{s_{i,t} + s_{i,t-1}}{2}. \quad (5)$$

Note that $\ln V$ in Equation (4) differs from $\ln V^{PF}$ in Equation (3): Equation (4) uses a translog aggregation of industry value-added growth, whereas Equation (3) uses the growth rate of value added simply aggregated across industries. The difference between the two is the reallocation of value added across industries.

3.4 Contribution of industry TFP growth and factor reallocation to aggregate TFP growth

To quantify the relative importance of individual industries and of factor reallocation in driving aggregate TFP growth, the direct-aggregation method of Jorgenson et al. (2007) is used. The aggregate production function is a value-added function, and aggregate value-added growth is measured as a translog index of industry value added, with weights equal to the industry share in aggregate nominal value added (as in Equation (4)). At the industry level, however, the production function is a gross-output (Y) function, so industry output growth can be decomposed into contributions from capital (K), labour (L) and intermediate input (X):

$$\Delta \ln Y_{i,t} = \bar{s}_{K,i,t} \Delta \ln K_{i,t} + \bar{s}_{L,i,t} \Delta \ln L_{i,t} + \bar{s}_{X,i,t} \Delta \ln X_{i,t} + \Delta \ln A_{i,t}. \quad (6)$$

In Equation (6), s_K , s_L and s_X are the shares of capital, labour and intermediate input in total nominal output of industry i ; $\Delta \ln Y_{i,t}$ is the growth rate of gross output of industry i in year t ; the term $\bar{s}_{K,i,t} \Delta \ln K_{i,t}$ is the contribution of capital growth, $\bar{s}_{L,i,t} \Delta \ln L_{i,t}$ the contribution of labour growth and $\bar{s}_{X,i,t} \Delta \ln X_{i,t}$ the contribution of intermediate-input growth; and $\Delta \ln A_{i,t}$ is the growth of industry total factor productivity.

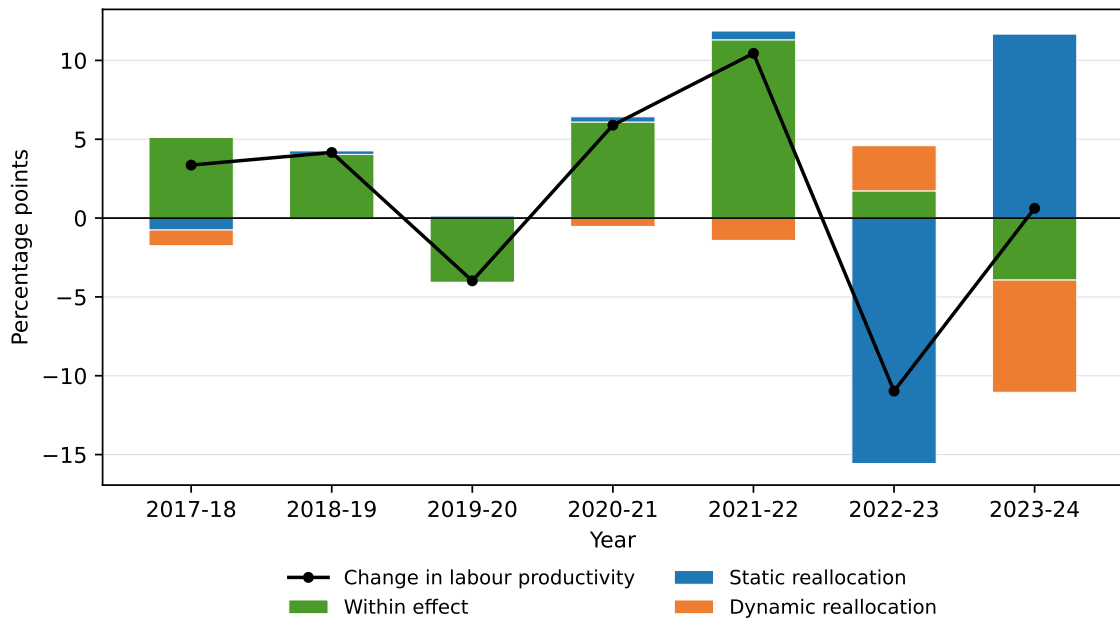
4 Results

4.1 Decomposition of labour-productivity growth

Computing aggregate labour productivity (Δv_t^{PF}) from the ASI data yields the decomposition reported in Table 1 and plotted in Figure 1.

Table 1: Decomposition of labour-productivity growth, Odisha manufacturing (percentage points)

Year	Within effect	Static reallocation	Dynamic reallocation	Change in labour productivity
2017–18	5.11384	−0.75619	−0.99769	3.35996
2018–19	4.037173	0.223073	−0.09996	4.160287
2019–20	−4.07705	0.126876	−0.02992	−3.98009
2020–21	6.092913	0.330518	−0.5303	5.893135
2021–22	11.30561	0.558412	−1.41655	10.44747
2022–23	1.725081	−15.5644	2.864366	−10.9749
2023–24	−3.93158	11.66825	−7.11745	0.619215

**Figure 1:** Decomposition of labour-productivity growth, Odisha manufacturing, 2017–18 to 2023–24. Stacked bars show the within, static and dynamic components; the line is the resulting change in labour productivity. Source: author’s computation from ASI data (values as in Table 1).

As Figure 1 shows, there is an extreme negative static reallocation in 2022–23. Three factors account for this:

1. **Post-pandemic recovery:** employment growth was absorbed by informal jobs, agriculture, self-employment and other sectors with lower value addition per worker.
2. **Post-COVID rebound distortions:** strong but uneven employment creation meant that many workers re-entered the workforce in low-skilled or service roles, lowering the aggregate allocative efficiency of the economy.
3. **Stagnant capital deepening:** lower capital intensity and sluggish investment in advanced machinery within certain industries meant that workers in growing sectors produced less output per hour worked.

The industry-wise analysis of the ASI data for the relevant periods shows that the major industries with negative static reallocation are those with the 3-digit NIC codes **192, 201, 210,**

241, **242** and **259**, representing, respectively, the manufacture of refined petroleum and LPG, basic chemicals and fertilisers, pharmaceuticals, basic iron and steel, non-ferrous metals and fabricated metal products.

- In the **refinery and LPG industry (192)**, 2022–23 was the peak of the Russia–Ukraine war, which caused global energy prices (crude oil, coking coal, natural gas) to skyrocket. While output value was squeezed by surging raw-material costs, these large, highly productive plants scaled back capacity utilisation or optimised labour to survive. Labour shifted from high-productivity upstream refining/smelting to lower-productivity fabrication, turning static reallocation negative (International Energy Agency, 2022).
- The **pharmaceuticals industry (210)** underwent automation and capital deepening; the surplus labour was absorbed by smaller, lower-tier, unorganised supply-chain units such as chemical mixing, low-tech plastic moulding and packaging.
- Global export demand collapsed sharply in late 2022 owing to rising worldwide interest rates. High-productivity export units in steel (**241**) and pharmaceuticals (**210**) faced underutilised capacity; because they retained workers despite lower global demand, productivity per worker fell, producing a negative static component.

The negative change in labour productivity in 2022–23 can be seen as an effect of the same forces. The 2022–23 labour market experienced what may be termed a “**low-unemployment, low-productivity trap**”: while the headline unemployment rate fell to 3.2% according to the Ministry of Statistics and Programme Implementation (2023) Annual PLFS Report, this employment expansion was heavily concentrated in the low-productivity self-employed segment — specifically own-account workers (OAWs) and unpaid family helpers, who exhibit significantly lower productivity than any category of paid worker. This amounted to a post-pandemic *reverse structural transformation*: instead of a smooth return to formal urban industries, the agricultural sector abnormally continued to absorb surplus labour, and the rise in its labour share dragged down aggregate allocative efficiency. Peak global inflation and intense geopolitical disruptions added to this by triggering energy and raw-material shocks that compressed the real gross value added of petro-chemical industries.

A similar pattern is visible in 2017–18, where the negative static reallocation reflected:

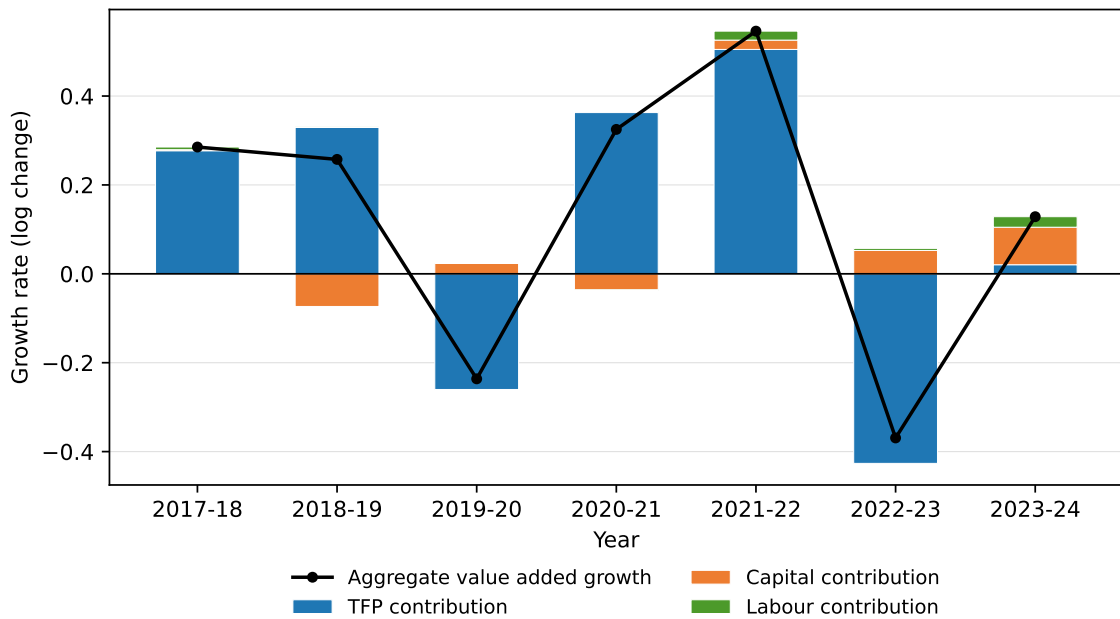
- twin regulatory shocks — demonetisation and the introduction of GST — which led industries to halt hiring;
- rapid informalisation of the manufacturing workforce — the informalisation of formal jobs through temporary, contractual or third-party workers instead of permanent payroll employment;
- premature de-industrialisation — labour bypassing manufacturing and moving from farming to low-productivity services such as transport, delivery and retail.

4.2 Contribution of factor inputs and TFP to value-added growth

Table 2 reports the growth-accounting decomposition of aggregate value-added growth (Equation (3)), and Figure 2 plots the same series. Aggregate TFP growth is the residual of aggregate value-added growth net of the aggregate growth of capital and labour inputs.

Table 2: Growth accounting for Odisha manufacturing (log growth rates)

Year	Growth of aggregate value added	Aggregate capital services growth rate	Aggregate labour input growth rate	Growth of aggregate TFP
2017–18	0.285283	0.00241	0.005953	0.27692
2018–19	0.25746	−0.07343	0.001455	0.329434
2019–20	−0.23627	0.02365	0.000291	−0.26021
2020–21	0.324934	−0.03598	−0.00185	0.362769
2021–22	0.546016	0.02094	0.020178	0.504897
2022–23	−0.36924	0.052618	0.004613	−0.42647
2023–24	0.128524	0.084417	0.023667	0.020439

**Figure 2:** Growth accounting decomposition of aggregate value added, Odisha manufacturing, 2017–18 to 2023–24. Stacked bars show the TFP, capital and labour contributions; the line is aggregate value-added growth. Source: author’s computation from ASI data (values as in Table 2).

The year-by-year pattern is instructive. In 2017–18 growth was driven purely by efficiency gains rather than by adding workers or machinery. In 2018–19 capital shrank, but large productivity gains more than offset this to yield strong total growth. In 2019–20 physical inputs stayed broadly flat, but a collapse in efficiency pulled total growth down. The year 2020–21 shows a classic post-crisis rebound: total inputs actually shrank, yet pure efficiency drove the entire expansion. The year 2021–22 was the best-performing year — while labour and capital grew slightly, TFP rose sharply and accounted for about 92% of overall economic expansion. In the immediately following year, however, despite expanding capital and hiring more workers, productivity collapsed and dragged down output; unlike the earlier efficiency-driven booms, the modest 2023–24 growth was powered primarily by heavy physical (capital) investment.

Interpretation of Figure 2. Figure 2 decomposes aggregate value-added growth into the contributions of total factor productivity (TFP), capital and labour for Odisha’s manufacturing sector. The line represents aggregate value-added growth, while the stacked bars show the

contributions of the three factors. TFP was the dominant contributor in most years: positive TFP contributions in 2017–18, 2018–19, 2020–21 and 2021–22 were associated with higher aggregate value-added growth, indicating that improvements in production efficiency, technology and resource utilisation were the primary drivers of manufacturing growth. By contrast, the contributions of capital and labour were relatively small throughout, implying that factor accumulation alone played a limited role. A marked decline is observed in 2019–20 and 2022–23, when aggregate value-added growth turned negative; this deterioration was driven primarily by negative TFP growth, despite only modest changes in capital and labour, implying a temporary decline in production efficiency rather than a substantial reduction in factor inputs. The recovery in 2023–24 reflects renewed positive contributions from both productivity and factor inputs. Overall, technological progress and efficiency improvements were the principal sources of manufacturing growth in Odisha, while capital and labour contributed only marginally—underscoring the importance of productivity-enhancing policies alongside investment in physical and human capital.

4.3 Industry gross-output growth

Table 4 reports the gross-output growth rates computed using Equation (6) from the ASI data for 2-digit industry divisions (NIC-2008) over 2014–15 to 2023–24, covering twenty-three manufacturing industries. Table 3 lists the corresponding NIC-2008 division descriptions.

Table 3: NIC-2008 2-digit manufacturing division descriptions

Code	Division	Code	Division
10	Food products	22	Rubber and plastic products
11	Beverages	23	Other non-metallic mineral products
12	Tobacco products	24	Basic metals
13	Textiles	25	Fabricated metal products, except machinery
14	Wearing apparel	26	Computer, electronic and optical products
15	Leather and related products	27	Electrical equipment
16	Wood and products of wood	28	Machinery and equipment n.e.c.
17	Paper and paper products	29	Motor vehicles, trailers and semi-trailers
18	Printing and reproduction of recorded media	30	Other transport equipment
19	Coke and refined petroleum products	31	Furniture
20	Chemicals and chemical products	32	Other manufacturing
21	Pharmaceuticals, medicinal chemical and botanical products	33	Repair and installation of machinery

Table 4: Gross-output growth rate by 2-digit NIC division, Odisha manufacturing (per cent)

NIC	2014–15	2015–16	2016–17	2017–18	2018–19	2019–20	2020–21	2021–22	2022–23	2023–24
All	59.12	72.26	70.75	71.55	79.51	87.20	79.21	65.18	87.88	105.70
1	27.37	−94.57	−243.20	−277.81	−713.33	−692.11	−86.44	96.02	164.11	194.45
10	148.67	127.70	168.07	153.64	91.32	111.07	105.95	119.52	131.46	137.70
11	29.17	46.54	51.68	−4.30	−6.26	34.02	32.46	33.98	34.84	36.06
12	9.16	15.12	21.60	21.19	18.73	17.95	17.27	27.15	38.31	29.79
13	89.24	82.80	76.19	54.89	46.40	44.86	28.46	31.54	46.96	45.26
14	14.95	13.54	12.42	80.50	25.11	0.86	74.19	29.03	12.38	18.96
16	99.07	449.60	411.60	58.39	163.94	172.43	−559.12	−527.01	285.75	241.16
17	45.46	52.02	41.28	33.58	30.29	26.18	25.32	26.72	26.87	27.44
18	51.21	41.60	55.90	69.89	61.85	48.98	43.10	33.85	31.13	33.37
19	73.71	56.80	21.05	75.82	116.85	605.54	564.61	173.28	138.68	154.92
20	57.40	119.81	103.22	27.66	13.23	24.21	29.57	55.70	69.86	74.91
21	13.28	19.35	31.32	33.01	39.03	24.09	23.97	26.24	18.84	19.10
22	185.24	151.65	158.29	193.82	239.32	123.18	78.03	91.21	71.28	70.50
23	31.57	41.63	40.05	35.63	162.17	141.45	−5066.27	−5076.67	55.03	57.48
24	52.18	65.81	67.73	64.38	57.28	101.71	63.93	50.08	81.04	99.84
25	80.17	104.47	108.20	75.19	56.87	40.02	57.28	101.18	117.37	90.92
27	193.37	181.34	186.02	212.93	105.04	128.37	206.74	219.65	203.06	137.41
28	51.65	66.96	57.59	38.41	72.79	32.89	26.35	25.31	28.15	39.83
29	49.89	56.99	80.83	150.37	169.98	46.39	62.64	57.24	73.32	61.31
30	12.52	47.81	71.00	58.93	23.47	21.04	21.99	31.06	42.07	46.50
31	29.79	43.71	66.41	−273.55	−286.23	98.64	71.19	56.98	83.07	79.74
33	27.66	59.31	13.88	11.63	22.88	5.08	38.54	54.86	54.66	51.45
Other	44.18	51.63	61.46	66.98	53.01	36.52	34.57	32.86	32.91	34.65

Note: Row labels are 2-digit NIC-2008 manufacturing divisions (see Table 3); “All” is the aggregate and “Other” groups the remaining divisions. The exact ASI category corresponding to code “1” should be confirmed by the author. A few series take extreme values (e.g. divisions 1, 16, 23, 31) in particular years, reflecting very small or restructured bases. Source: author’s computation from ASI data.

Interpretation of Table 4. Gross-output growth reflects the combined effect of changes in production, demand, investment and technological improvement. At the aggregate level the manufacturing sector exhibited an upward trend, rising from 59.12 per cent in 2014–15 to 105.70 per cent in 2023–24, though with considerable year-to-year fluctuation. Industry-level growth was highly heterogeneous. Several industries—food products (NIC 10), chemicals (NIC 20), rubber and plastics (NIC 22), fabricated metal products (NIC 25), electrical equipment (NIC 27) and motor vehicles (NIC 29)—recorded positive growth through most of the period and appear to have been the principal contributors to manufacturing expansion. By contrast, tobacco products (NIC 12), beverages (NIC 11), wood products (NIC 16) and repair and installation of machinery (NIC 33) displayed sharp fluctuations, reflecting changes in demand, raw-material availability, investment and capacity. The wood products industry (NIC 16) recorded exceptionally high growth in 2015–16 and 2016–17, a sharp contraction in 2020–21 and 2021–22, and a recovery thereafter, while the non-metallic mineral products industry (NIC 23) experienced an unprecedented decline during the pandemic years. Overall, manufacturing growth in Odisha was uneven across industries, with a few industries emerging as major growth drivers and others fluctuating in response to economic shocks and structural adjustments.

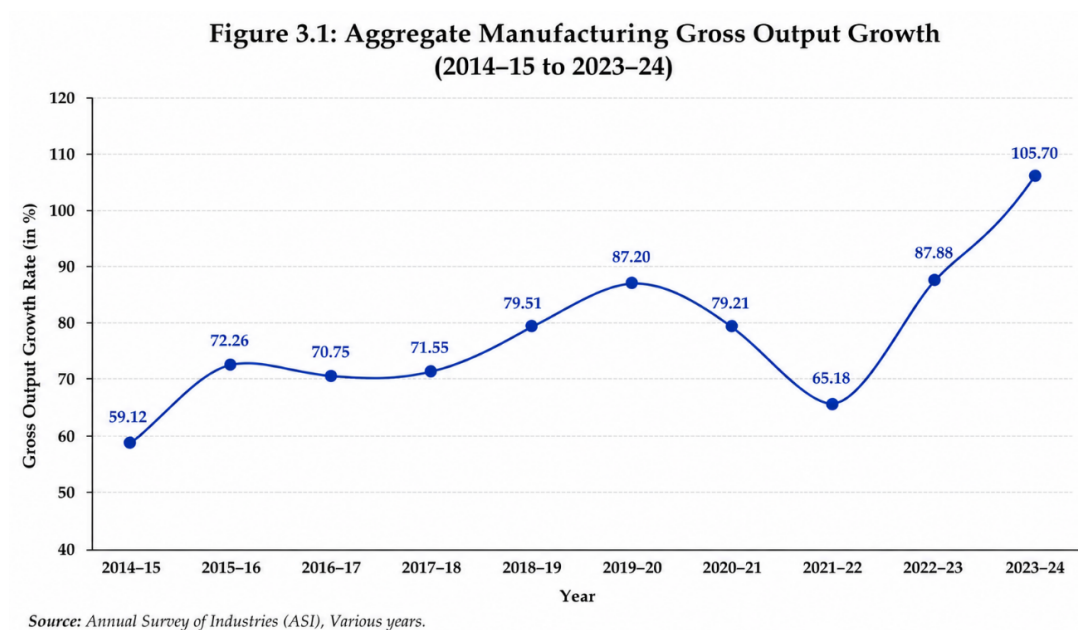
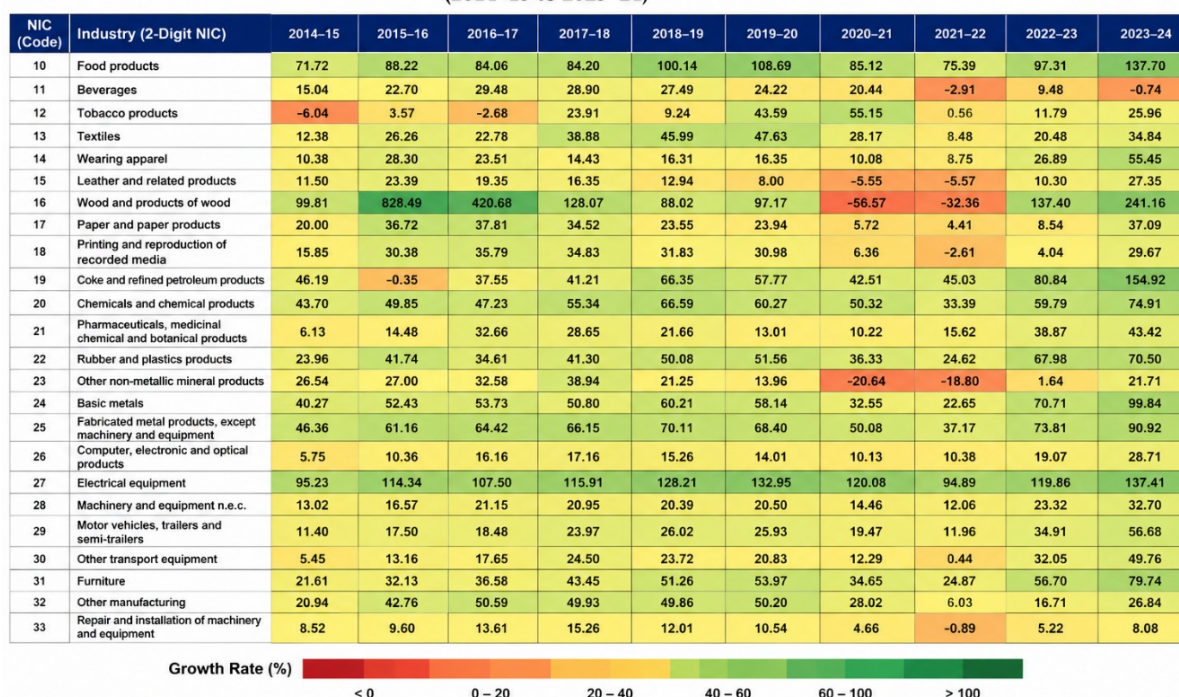


Figure 3: Aggregate manufacturing gross-output growth, Odisha, 2014–15 to 2023–24. Source: Annual Survey of Industries (ASI), various years.

Figure 3 illustrates the trend in aggregate gross-output growth. The overall pattern is one of long-term expansion despite short-term fluctuation: growth rose from about 59.12 in 2014–15 to 105.70 in 2023–24. It remained relatively stable up to 2019–20, moderated during 2020–21 and 2021–22 (reflecting COVID-19 disruptions, supply-chain bottlenecks and weaker industrial demand), then recovered strongly in 2022–23 and reached its highest level in 2023–24, indicating improved industrial production, increased investment and stronger demand.

**Figure 3.2: Heat Map of Gross Output Growth Rate by 2-Digit NIC Industries
(2014–15 to 2023–24)**

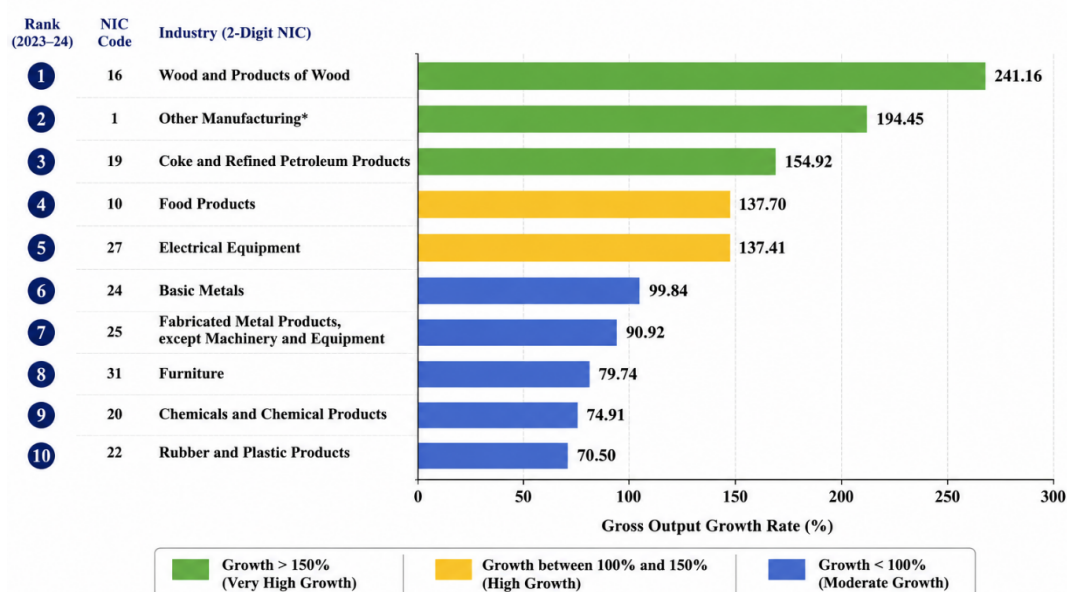


Source: Annual Survey of Industries (ASI), Various years.

Figure 4: Heat map of gross-output growth rate by 2-digit NIC industry, Odisha, 2014–15 to 2023–24. Darker green denotes higher growth; orange to red denotes low or negative growth. Source: Annual Survey of Industries (ASI), various years.

The heat map in Figure 4 summarises industry-wise growth patterns and confirms their heterogeneity. Industries such as food products (NIC 10), electrical equipment (NIC 27), fabricated metal products (NIC 25), basic metals (NIC 24), chemicals (NIC 20) and rubber and plastic products (NIC 22) recorded moderate-to-high growth throughout, whereas wood products (NIC 16), other non-metallic mineral products (NIC 23), tobacco products (NIC 12) and repair and installation of machinery (NIC 33) shifted frequently between high and low growth. The effects of the COVID-19 pandemic are visible in 2020–21 and 2021–22, when several industries recorded declines, followed by a broad-based recovery in 2022–23 and 2023–24 led by electrical equipment, food products, basic metals, fabricated metals and motor vehicles.

Figure 3.3: Top 10 Manufacturing Industries by Gross Output Growth Rate in Odisha (2023–24)



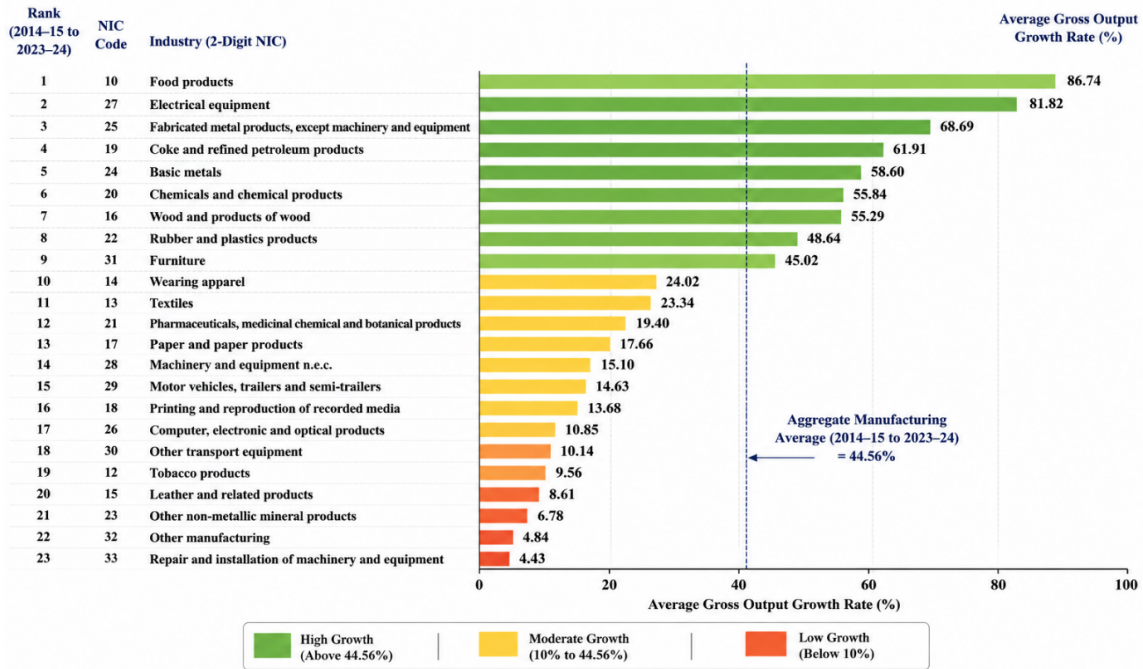
Note: * Please refer to your dataset for the exact industry name corresponding to NIC Code 1.

Source: Computed from Annual Survey of Industries (ASI), 2023–24.

Figure 5: Top 10 manufacturing industries by gross-output growth rate, Odisha, 2023–24. Source: computed from Annual Survey of Industries (ASI), 2023–24.

Figure 5 ranks the ten fastest-growing industries in 2023–24. Wood and products of wood (NIC 16) recorded the highest growth (241.16 per cent), followed by NIC code 1 (194.45 per cent) and coke and refined petroleum products (NIC 19, 154.92 per cent); these fall in the very-high-growth category (above 150 per cent). Food products (NIC 10, 137.70 per cent) and electrical equipment (NIC 27, 137.41 per cent) lie in the high-growth category (100–150 per cent), while basic metals (NIC 24), fabricated metal products (NIC 25), furniture (NIC 31), chemicals (NIC 20) and rubber and plastic products (NIC 22) recorded moderate growth between 70.50 and 99.84 per cent. The concentration of exceptionally high growth in only a few industries implies that aggregate manufacturing performance depends significantly on a limited number of key industries.

Figure 3.4: Average Gross Output Growth Rate by 2-Digit NIC Industries in Odisha (2014–15 to 2023–24)



Note: Average growth rate is the arithmetic mean of annual gross output growth rates from 2014–15 to 2023–24.
Source: Computed from Annual Survey of Industries (ASI), Various years.

Figure 6: Average gross-output growth rate by 2-digit NIC industry, Odisha, 2014–15 to 2023–24; the vertical line marks the aggregate manufacturing average of 44.56 per cent. Source: computed from Annual Survey of Industries (ASI), various years.

Figure 6 ranks industries by average gross-output growth over the full period against the aggregate manufacturing average of 44.56 per cent. Food products (NIC 10) recorded the highest average growth (86.74 per cent), followed by electrical equipment (NIC 27, 81.82 per cent) and fabricated metal products (NIC 25, 68.69 per cent). Coke and refined petroleum products (NIC 19), basic metals (NIC 24), chemicals (NIC 20), wood products (NIC 16), rubber and plastics (NIC 22) and furniture (NIC 31) also exceeded the aggregate average, while wearing apparel (NIC 14), textiles (NIC 13), paper products (NIC 17), machinery (NIC 28), motor vehicles (NIC 29), computer and electronic products (NIC 26), other manufacturing (NIC 32) and repair and installation (NIC 33) remained below it. Long-term manufacturing growth in Odisha was thus concentrated in a relatively small number of industries, highlighting the unequal pace of structural transformation across the sector.

5 Discussion

Four findings emerge consistently from the results above. First, the *within-industry* effect is the dominant source of labour-productivity change in Odisha manufacturing in most years (Table 1): in 2017–18, 2018–19, 2020–21 and 2021–22 it is large and positive, and it is the main driver of the corresponding rise in aggregate labour productivity. Reallocation effects — static and dynamic — are usually an order of magnitude smaller, indicating that productivity growth has come mainly from upgrading *within* industries rather than from shifting workers *between* them.

Second, when reallocation does matter, it can dominate. The 2022–23 episode is the clearest case: a static reallocation of –15.56 percentage points overwhelms a positive within effect of 1.73 and turns the change in labour productivity sharply negative (–10.97). The interpretation offered in Section 4 — post-pandemic reverse structural transformation, informalisation,

retention of labour in high-productivity export industries facing weak demand, and energy-price shocks in 3-digit petro-chemical and metal industries — is consistent with a “low-unemployment, low-productivity trap” in which rising employment coincides with falling allocative efficiency. The partial reversal in 2023–24, when static reallocation rebounds to +11.67 while the dynamic term falls to −7.12, is consistent with a subsequent re-sorting of labour across industries.

Third, the growth-accounting results (Table 2) reinforce the within-industry story at the aggregate level: total factor productivity, not factor accumulation, accounts for most of the year-to-year movement in aggregate value added. Capital-services and labour-input growth are small in absolute terms in almost every year, so aggregate value-added growth and TFP growth move closely together. The exception is 2023–24, when capital and labour grew comparatively strongly but TFP growth nearly vanished (0.0204), leaving a modest, input-driven expansion — a qualitatively different growth pattern from the efficiency-driven booms of the preceding years.

Fourth, the gross-output evidence (Table 4 and Figures 3–6) shows that aggregate manufacturing expansion masks pronounced heterogeneity across industries. A small group of industries — notably food products, electrical equipment, fabricated and basic metals, chemicals and rubber and plastics — consistently outperformed the sector average and drove the recovery of 2022–23 and 2023–24, while others fluctuated sharply with demand and investment cycles. This concentration implies that aggregate manufacturing performance, and hence the aggregate TFP and reallocation results, depend heavily on a few key industries. Because Equation (3) is expressed in log differences while Equation (2) is expressed in levels, the two decompositions are complementary rather than directly comparable in magnitude, but they tell a mutually consistent story.

6 Conclusion

This study examined the sources of manufacturing growth and labour productivity in Odisha using industry-level ASI data for the period 2017–18 to 2023–24. By employing growth-accounting and labour-productivity decomposition techniques, it investigated the relative contributions of capital, labour, intermediate inputs, total factor productivity (TFP) and structural change to aggregate manufacturing performance.

The analysis revealed that Odisha’s manufacturing sector experienced significant fluctuations in output and productivity during the study period. While aggregate gross output exhibited an overall upward trend, the pace of growth differed considerably across industries, reflecting the heterogeneous nature of manufacturing development. Industries such as food products, electrical equipment, fabricated metal products, basic metals and chemicals consistently recorded higher growth and emerged as important drivers of manufacturing expansion.

The growth-accounting results indicate that total factor productivity was the principal source of aggregate value-added growth in most years, suggesting that improvements in technology, production efficiency and organisational practices contributed more to manufacturing growth than increases in factor inputs alone. Although capital and labour contributed positively, their contributions remained relatively modest compared with productivity improvements. The sharp decline in TFP during certain years, particularly around the post-pandemic period, also highlights the vulnerability of manufacturing performance to external economic shocks.

The labour-productivity decomposition further shows that aggregate productivity growth was driven primarily by improvements within industries rather than by the reallocation of labour across industries. The within-industry effect consistently contributed the largest share of productivity growth, while the static and dynamic reallocation effects displayed substantial year-to-year variation. The pronounced negative static reallocation in 2022–23 indicates that

labour shifted towards relatively less productive industries during the post-pandemic recovery, temporarily reducing aggregate productivity growth.

Overall, the findings suggest that sustained manufacturing growth in Odisha depends not only on expanding capital investment but also on improving production efficiency, encouraging technological adoption and facilitating the movement of resources towards more productive industries. The study provides empirical evidence on the role of structural change in manufacturing productivity and contributes to a better understanding of the factors influencing industrial development in Odisha. The analytical framework adopted here can also serve as a useful reference for future research on manufacturing productivity and structural transformation in other Indian states.

7 Policy Implications

The findings have several policy implications for strengthening manufacturing productivity and promoting sustainable industrial development in Odisha.

First, since total factor productivity emerged as the major contributor to manufacturing growth, policies should prioritise technological upgrading, digital transformation and innovation across manufacturing industries. Initiatives that encourage automation, the adoption of advanced production technologies and investment in research and development can significantly enhance production efficiency and long-term competitiveness.

Second, the dominance of within-industry productivity improvements suggests that raising efficiency within existing industries should remain a key policy objective. This can be achieved through modernisation of production processes, improved managerial practices, energy-efficient technologies and workforce skill development, with special emphasis on micro, small and medium enterprises (MSMEs), which often face constraints in adopting modern technologies.

Third, the negative static reallocation observed during the study period indicates that labour was not always moving towards more productive industries. Policies should therefore facilitate efficient resource allocation by improving labour mobility, strengthening vocational training and aligning skill-development programmes with the requirements of high-productivity sectors, while encouraging the expansion of consistently higher-productivity industries.

Fourth, the considerable variation in growth across industries calls for sector-specific industrial strategies. High-performing industries such as food products, electrical equipment, fabricated metal products, basic metals and chemicals should receive continued support through infrastructure development, investment promotion, improved logistics and stronger linkages with domestic and global value chains, while persistently low-growth industries should be supported through technology adoption, access to finance and capacity-building.

Finally, strengthening industrial infrastructure, ensuring reliable power supply, expanding transport connectivity and improving the overall business environment would help attract private investment and enhance manufacturing competitiveness. Continuous monitoring of productivity indicators using ASI data can assist policymakers in designing evidence-based policies that promote productivity-led and structurally balanced manufacturing growth in Odisha.

8 Limitations and Future Work

Several limitations should be borne in mind. First, the study period is short (about a decade) and includes the unusual COVID-19 disruption, so single-year reallocation estimates — particularly for 2020–21 to 2022–23 — should be read with caution. Second, the ASI covers only the

formal (organised) manufacturing sector; because much of the post-pandemic employment shift was towards informal and own-account activity, the decompositions capture the *consequences* of informalisation for formal-sector allocative efficiency but not the informal sector itself. Third, a few 2-digit and 3-digit series take extreme values in particular years (e.g. NIC divisions 1, 16, 23 and 31 in Table 4), reflecting very small or restructured bases; such observations can heavily influence aggregates and warrant separate scrutiny. Fourth, as noted in Section 3, the level-based and log-based decompositions are not strictly comparable in magnitude, and the results depend on the official deflators and factor-share measures used.

Future work could extend the analysis in several directions: lengthening the panel as more ASI rounds become available; incorporating the unorganised manufacturing sector to obtain a fuller picture of reallocation; disaggregating to the district level within Odisha; benchmarking Odisha against other Indian states; and examining the roles of capital deepening, skill composition and trade exposure in shaping the within-industry and reallocation components. Robustness checks using alternative deflation and aggregation methods would further strengthen the conclusions.

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